Lecture-11
Inviscid Flows II: Bernoulli Equation

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Learning Objectives

By the end of this lecture, you should be able to

1. Contrast the physical implications of deriving the **Bernoulli equation** along a single streamline versus deriving it in Cartesian coordinates
2. Distinguish between **static, dynamic, and stagnation** pressure.
3. Define the **stagnation state** of a moving fluid particle

Bernoulli Equation

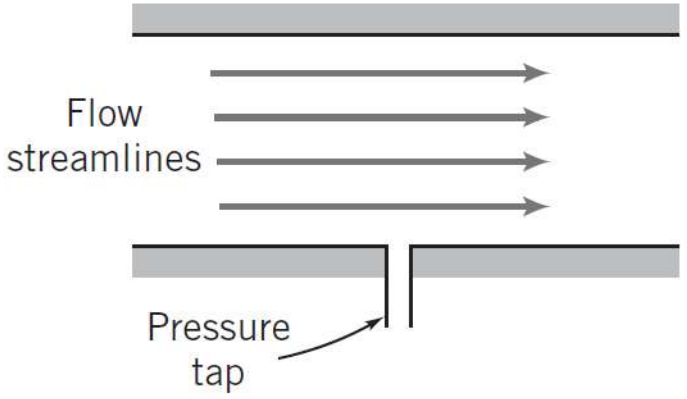
- Euler's equation, which is basically a law for conserving momentum of a fluid element, is **difficult to solve analytically** except for some basic flow problems.
- An approach to yield a simplified equation is to **integrate Euler's equation along a streamline**. The resulting equation is called the **Bernoulli Equation**.

$$\int \frac{dp}{\rho} + \frac{V^2}{2} + gz = \text{constant} \qquad (\text{along } s)$$

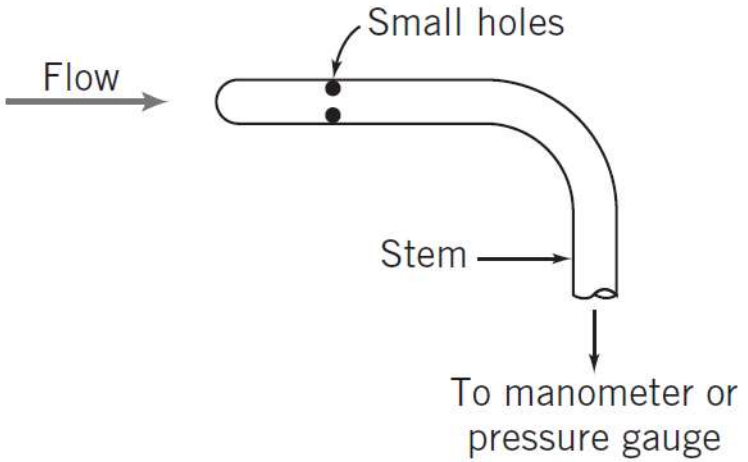
Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Static, Stagnation and Dynamic Pressures

- The pressure we used in deriving the Bernoulli Equation is commonly called the **static pressure**.
- The **stagnation pressure** is obtained when a flowing liquid is **decelerated to rest** by a frictionless process.
- The **difference between the stagnation and static pressures**, which is a result of the motion of the fluid, is referred to as the **dynamic pressure**.



(*a*) Wall pressure tap



(*b*) Static pressure probe

$$p_0 = p + \frac{1}{2}\rho V^2$$

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Revisiting the learning objectives

1. Contrast the physical implications of deriving the **Bernoulli equation** along a single streamline versus deriving it in Cartesian coordinates
2. Distinguish between **static, dynamic, and stagnation** pressure.
3. Define the **stagnation state** of a moving fluid particle

